Rishi Ghosh

Email | GitHub | LinkedIn | Portfolio | X / Twitter

Professional Summary

Open-source contributor, full-stack developer, and AI builder pursuing a B.Tech in Computer Science and Engineering (Cyber Security). Focused on building scalable software, exploring emerging technologies, and contributing to developer-focused projects and the open-source ecosystem. Experienced in developing React and Next.js applications, integrating REST APIs, building AI-powered applications, and working with modern JavaScript and Python technologies. Passionate about collaborative software development, Git-based workflows, problem-solving, and creating practical solutions through code.

Experience

Co-Founder, COO & CTO Dec 2025 – Aug 2026
Foretyx AI – Enterprise LLM Security Platform *Remote*

- Architected a production React, Vite, and TypeScript admin panel for an AI security gateway, enabling real-time LLM request monitoring, personally identifiable information (PII) tracking, and policy enforcement with a sub-200ms SLA.
- Developed audit-log, incident management, and security policy modules using Zustand state management; owned JWT authentication flows and frontend-backend REST API integration.
- Implemented Docker-based deployment and GitHub Actions CI/CD workflows; contributed to technical leadership, engineering collaboration, and ML researcher onboarding within the founding team.

Founder & Full Stack Developer June 2026 – Present
Agnio – Web Development Agency *Remote*

- Founded and operate a web development agency delivering custom web applications and responsive websites using React, Next.js, TypeScript, and modern frontend technologies.
- Manage end-to-end development workflows, including UI/UX implementation, frontend architecture, API integration, deployment, client communication, and project delivery.
- Build production-oriented web experiences using Tailwind CSS, reusable components, Git-based version control, and Vercel deployment pipelines.

Full Stack Developer Aug 2026 – Oct 2026
Cosmodex *Remote*

- Improved an interactive learning platform through frontend development, UI enhancements, navigation improvements, mascot interactions, and reusable component development.
- Developed Python Galaxy, an interactive educational experience featuring eight learning stations and Monaco Editor integration.
- Resolved frontend bugs, improved TypeScript type safety, and collaborated on software development workflows using Git and GitHub.

Open Source & Technical Projects

ResearchOS – AI Research Agent | *LangGraph, FastAPI, Groq, React* | GitHub | Live Demo 2025

- Designed a multi-agent research system using LangGraph and Groq's Llama 4 Scout, implementing a four-agent workflow consisting of Search, Reader, Writer, and Critic agents.
- Developed a React and Vite dashboard featuring live agent execution tracking, real-time terminal logs, and Tavily API integration for web search and research retrieval.
- Built backend services using FastAPI and integrated LLM-powered research and automated report generation, reducing research and report generation time to under 30 seconds.

Foretyx Secure Gateway | *React, Vite, TypeScript, Zustand, Docker* | Website 2024 – 2026

- Developed a production admin panel for AI security middleware, implementing security policy configuration, audit-log exploration, PII monitoring, and LLM gateway controls.
- Implemented reusable frontend components, centralized state management, authentication workflows, and API integration for an enterprise-oriented AI security platform.

Lume – Codebase Security Visualizer | *Next.js 14, D3.js, Supabase, Hugging Face* | GitHub 2025

- Built a D3.js-powered force-directed codebase visualization tool mapping security debt and OWASP/CWE vulnerabilities into interactive blast-radius risk zones across repositories.
- Integrated a Hugging Face Mistral LLM to generate automated executive summaries and AutoFix patches, featuring live Git diff previews.

Technical Skills

Programming Languages: JavaScript (ES2023+), TypeScript, Python, HTML5, CSS3
Open Source & Developer Tools: Git, GitHub, Version Control, Git Workflows, Repository Management, Collaborative Development, GitHub Actions
Frontend Development: React.js, Next.js 14, Vite, Tailwind CSS, Zustand, React Router, Shadcn/UI, D3.js, Responsive Web Design, Component-Based Architecture, UI/UX Implementation
Backend Development & APIs: FastAPI, REST APIs, API Integration, JWT Authentication, Backend-Frontend Integration, HTTP, JSON
AI / Machine Learning: Large Language Models (LLMs), LangGraph, LangChain, Multi-Agent Systems, Groq, OpenRouter, Hugging Face, Prompt Engineering, AI Application Development, LLM API Integration
Databases & Cloud: Supabase, PostgreSQL, Database Integration, Vercel, Docker, CI/CD, Deployment Automation
Software Engineering: Data Structures and Algorithms, Object-Oriented Programming, Debugging, Problem Solving, TypeScript Type Safety, Software Development Lifecycle

Education

Netaji Subhash Engineering College Kolkata, India
B.Tech in Computer Science & Engineering (Cyber Security) *Sep 2025 – Present*